
ANALYZING MEMORIZATION IN VAR-D16 WITH UNITMEM

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ABSTRACT

We apply UnitMem to pretrained VAR-d16 to locate its highest-scoring units and identify the training images that activate them most selectively. Using 10,000 ImageNet training images and ten augmentations per image, we compute UnitMem at each image-dependent generation scale and average the scores across scales. Of the top 10% of units, 92.1% are located in fc1 layers and 56.9% occur in blocks 0–2. For each selected unit, we identify its most activating training image at each scale. These responses concentrate on a small subset of the data: 3.8% of images account for half of all selections. Compared with class-matched controls, the most frequently selected images have higher Laplacian sharpness and require more JPEG bytes per pixel.

1 INTRODUCTION

VAR generates images autoregressively from coarse to fine scales (Tian et al., 2024). Its Transformer reuses the same blocks at each generation scale. UnitMem, originally proposed for self-supervised vision encoders, quantifies a unit’s image selectivity by comparing its response to the most activating image with its mean response to all others (Wang et al., 2024).

Kasliwal et al. adapted UnitMem to VAR by computing fc1 scores separately at each generation scale (Kasliwal et al., 2025). Building on this formulation, we ask: (1) which layer types and block depths contain the highest-scoring 10% of units; and (2) which training images activate these units most strongly, and what properties distinguish them from class-matched controls?

2 METHOD

We evaluate pretrained VAR-d16 on a balanced subset of 10,000 ImageNet training images (10 per class) (Russakovsky et al., 2015), using teacher forcing and ten augmentations per image. Following the scale-wise VAR formulation of UnitMem (Wang et al., 2024; Kasliwal et al., 2025), the response of unit u to image x at scale s is its mean absolute activation over views and tokens:

$$a_{u,s}(x) = \frac{1}{10} \sum_{r=1}^{10} \frac{1}{|T_s|} \sum_{t \in T_s} |h_{u,t}(x^{(r)})|.$$

UnitMem is computed independently at each scale by comparing the most activating image with the mean response to the remaining images in D' :

$$\text{UnitMem}_{u,s} = \frac{\mu_{\max,u,s} - \mu_{-\max,u,s}}{\mu_{\max,u,s} + \mu_{-\max,u,s}}.$$

RQ1. We average the nine image-dependent scale scores, $S_u = \frac{1}{9} \sum_{s=1}^9 \text{UnitMem}_{u,s}$, and jointly rank fc1, attn.proj, and fc2 units from all 16 blocks. Scale 0 is excluded because it contains only the class condition. We select the overall top 10% and report the fraction of each layer type and block entering this set, together with the joint layer-type-by-block distribution.

RQ2. For each selected unit and scale, we record $x_{u,s}^* = \arg \max_{x \in D'} a_{u,s}(x)$. Each unit-scale pair casts one vote, and images are ranked by their total votes. We compare the top-50 images with class-matched controls on eight predefined low-level features: grayscale entropy, edge density, Laplacian sharpness, luminance contrast, mean saturation, colorfulness, center-to-border contrast,

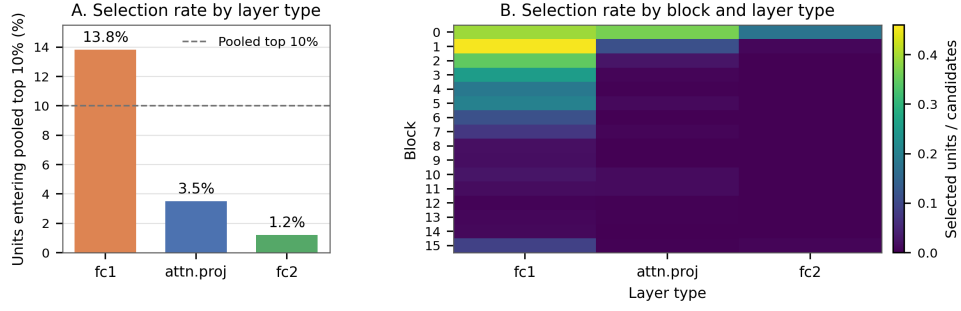


Figure 1: Location of the overall top 10% UnitMem units in VAR-d16: **A** fraction of each layer type entering the pooled top decile; **B** selection rate for each layer-type–block pair.

and JPEG bytes per pixel. We blindly annotated seven predefined patterns in a randomized contact sheet of the top-50 images and 50 class-matched controls, with labels and votes hidden. Class-matched permutation tests (class-stratified for annotations) and separate BH-FDR corrections were used. Cohen’s d and h measure effect sizes for continuous features and binary patterns, respectively; q denotes the BH-FDR-adjusted p -value.

3 RQ1: WHERE ARE THE HIGHEST-SCORING UNITS?

Layer type. Fc1, attn.proj, and fc2 selection rates are 13.82%, 3.51%, and 1.21%; fc1 therefore accounts for 92.14% of the 9,830 selected units.

Block location. Blocks 0–2 contain 56.93% of selected units, with the highest rate in block 1’s fc1 layer (45.43%). Rates generally decrease with depth, followed by a smaller rise in block 15.

4 RQ2: WHICH IMAGES ARE SELECTED?

Vote distribution. Of 10,000 images, 6,319 (63.19%) receive a vote, while 383 (3.83%) account for half. The top 10/50 receive 9.39%/21.69% of all votes. The most-selected image receives 2,188 unit-scale votes cast by 464 distinct units across 15 blocks, three layer types, and eight scales.

Image properties. Only Laplacian sharpness ($d = 0.64$, $q = 0.0008$) and JPEG bytes per pixel ($d = 0.43$, $q = 0.0072$) remain significant after BH-FDR correction; the other six features do not (all $q \geq 0.135$).

Semantic patterns. Repetitive textures/geometric patterns (56% vs. 28%; $h=0.58$, $q=0.012$) and extreme close-ups/crops (46% vs. 22%; $h=0.51$, $q=0.031$) were the only patterns enriched after BH-FDR; the other five showed no reliable differences.

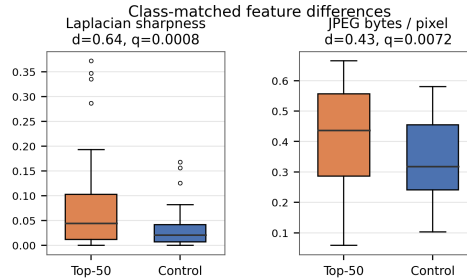


Figure 2: RQ2 image features that differ from class-matched controls after BH-FDR correction.

5 CONCLUSION

Equal averaging across the nine image-dependent scale scores places high-UnitMem units mainly in early fc1 layers (92.1% fc1; 56.9% blocks 0–2). Selections are concentrated (3.8% of images receive half) and favor images with higher Laplacian sharpness, more JPEG bytes per pixel, repetitive textures, and extreme close-ups relative to class-matched controls.

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